CHANDIGARH UNIVERSITY

APEX INSTITUTE OF TECHNOLOGY

ASSIGNMENT 1

Subject: Computer Networks
Subject Code: 24CST-339

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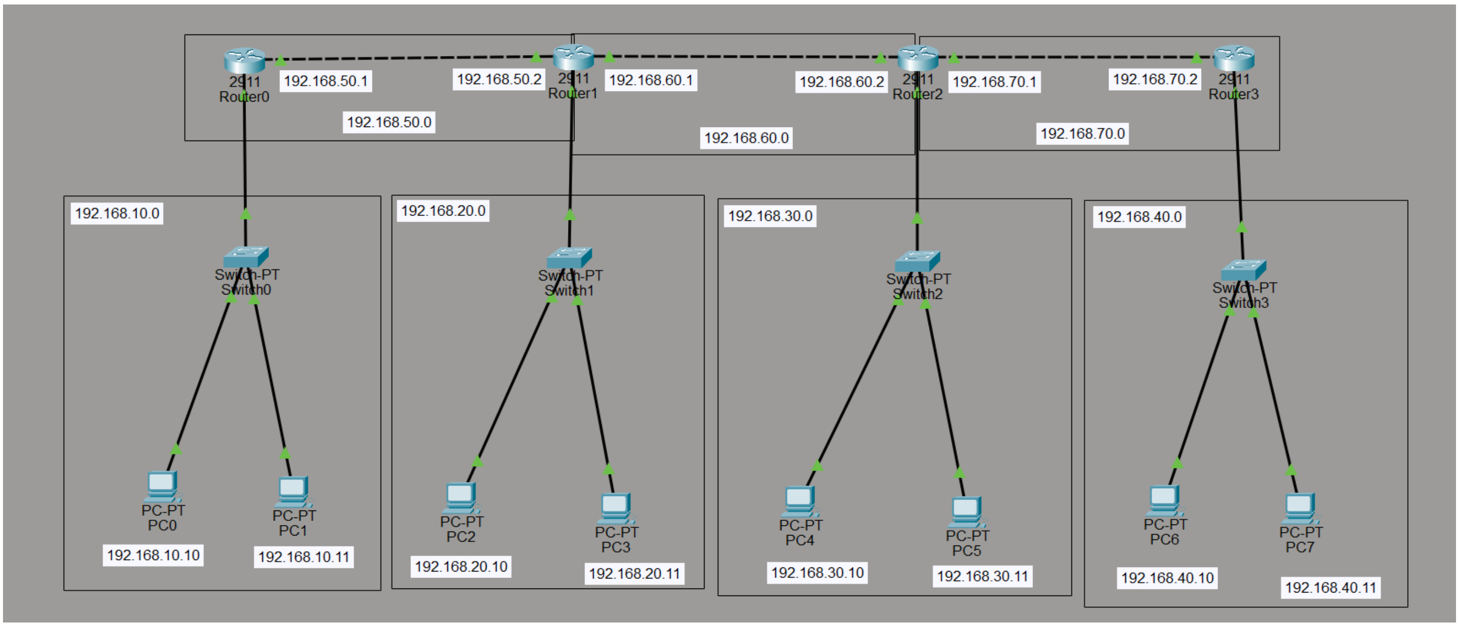
Question 1.

Develop Static Routing with 4 routers, 4 Switches G 8 Pc's. Make sure to highlight the Networks with their network ID's. IP Addresses should be written in text format as well.



Answer:

Static Routing Table:

Router	Destination Network	Subnet Mask	Next Hop
Router0	192.168.20.0	255.255.255.0	192.168.50.2
	192.168.30.0	255.255.255.0	192.168.50.2
	192.168.40.0	255.255.255.0	192.168.50.2
Router1	192.168.10.0	255.255.255.0	192.168.50.1
	192.168.30.0	255.255.255.0	192.168.60.2
	192.168.40.0	255.255.255.0	192.168.60.2
Router2	192.168.10.0	255.255.255.0	192.168.60.1
	192.168.20.0	255.255.255.0	192.168.60.1
	192.168.40.0	255.255.255.0	192.168.70.2
Router3	192.168.10.0	255.255.255.0	192.168.70.1
	192.168.20.0	255.255.255.0	192.168.70.1
	192.168.30.0	255.255.255.0	192.168.70.1

Network Table:

Network ID	Purpose	Devices Connected
192.168.10.0	LAN	Router0, PC0, PC1
192.168.20.0	LAN	Router1, PC2, PC3
192.168.30.0	LAN	Router2, PC4, PC5
192.168.40.0	LAN	Router3, PC6, PC7
192.168.50.0	WAN	Router0 to Router1
192.168.60.0	WAN	Router1 to Router2
192.168.70.0	WAN	Router2 to Router3

Question 2:

Write the difference between Hub, Switch & Router in Tabular Format. Explain why Router is the most intelligent and Hub is the least intelligent device.

Answer:

Networking devices are used to connect computers and other devices in a network. Hub, Switch, and Router are three fundamental networking devices that differ greatly in how they handle and forward data.

Difference between Hub, Switch and Router:

Parameter	**Hub**	**Switch**	**Router**
OSI Layer	Physical Layer (Layer 1)	Data Link Layer (Layer 2)	Network Layer (Layer 3)
Addressing	Does not use any address	Works with MAC addresses	Works with IP addresses
Forwarding Method	Sends data to every connected port	Delivers data only to the intended port	Forwards packets to the correct network
Intelligence	None — no decision making	Basic — learns MAC addresses	High — maintains and processes routing tables
Collision Domain	Single shared collision domain	Separate collision domain per port	Separate collision domain per interface
Broadcast Domain	One for all ports	One for all ports (unless VLANs used)	Separate broadcast domain per interface
Duplex Mode	Half-duplex only	Full-duplex supported	Full-duplex supported
Network Filtering	Not possible	Filters at MAC level	Filters at IP level using ACLs
Connects	Devices within same LAN	Devices within same LAN	Different networks and subnets
Security Level	Very low	Medium	High
Traffic Generated	High — unnecessary broadcasts	Low — targeted delivery	Minimal — only forwards what is needed
Cost	Very low	Medium	High

Why is Router the Most Intelligent Device?

A router is considered the most intelligent networking device because:

- It works at the Network Layer (Layer 3) of the OSI model.
- It uses IP addresses to identify source and destination networks.
- It maintains a Routing Table to determine the best path for data transmission.
- It connects different networks such as LANs and WANs.
- It supports dynamic routing protocols (RIP, OSPF, BGP) to automatically update routes.
- It reduces unnecessary traffic by forwarding packets only to the correct network.
- It provides advanced features like NAT, ACLs, QoS, and VPN support.

Why is Hub the Least Intelligent Device?

A hub is considered the least intelligent networking device because:

- It works at the Physical Layer (Layer 1) of the OSI model.
- It does not understand MAC addresses or IP addresses.
- It simply broadcasts data to all connected devices, regardless of the destination.
- It cannot make any forwarding decisions.
- It creates a single collision domain — when two devices transmit simultaneously, a collision occurs.
- It only supports half-duplex communication (cannot send and receive at the same time).
- It creates unnecessary network traffic and increases the chance of collisions.

A Hub is a basic device that blindly broadcasts data to all devices, a Switch forwards data intelligently using MAC addresses within a LAN and a Router connects different networks using IP addresses and selects the best path for data transmission. Therefore, the Router is the most intelligent, while the Hub is the least intelligent networking device.

Question 3:

What are Random Access Protocols? Why do we need them? Explain each category with diagram and example. Also explain the difference between CSMA/CD and CSMA/CA.

Answer:

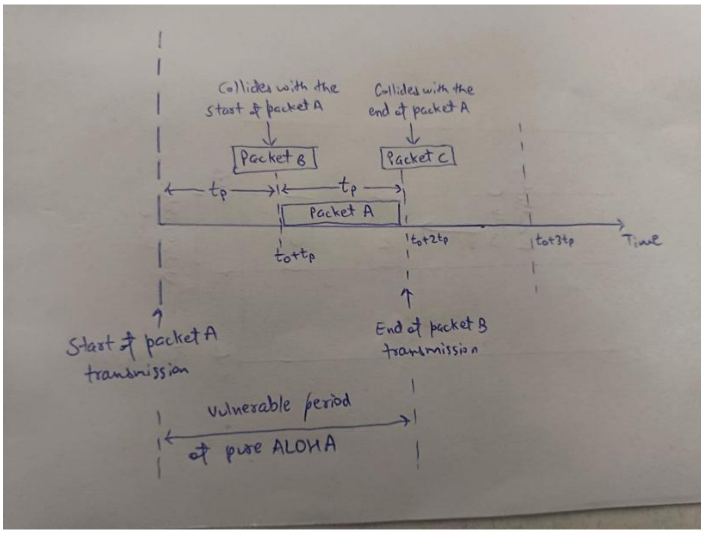
In modern computer networks, a single shared communication medium is often accessed by multiple devices simultaneously. Without any coordination mechanism, two or more devices may transmit at the exact same moment — causing their signals to interfere with each other. This interference is called a collision, and it results in corrupted or lost data.
Random Access Protocols are a set of MAC (Medium Access Control) layer rules that govern how devices compete for and share a common communication channel. Unlike controlled access methods (which require a central authority to assign turns), random access protocols allow each device to independently decide when to transmit — making them decentralized and scalable.

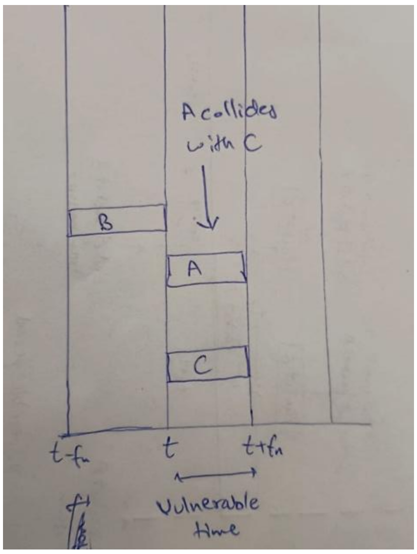
Why Do We Need Random Access Protocols?

- Multiple devices need to share a single channel — there must be a fair and organized way to do this.
- Without these protocols, uncontrolled simultaneous transmissions would cause constant collisions and make the network unusable.
- They eliminate the need for a central controller, making the network more robust and scalable.
- They define clear rules for what a device should do when a collision occurs — ensuring data is eventually delivered.
- They maximize channel utilization while keeping collision rates acceptably low.

Categories of Random Access Protocols:

1. ALOHA
ALOHA is the earliest and most basic random access protocol, developed at the University of Hawaii for wireless packet radio networks. It operates on a simple principle — transmit whenever you have data, and deal with collisions after they happen.
Pure ALOHA:
Any station can transmit data at any time without checking whether the channel is occupied. After transmitting, the sender waits for an acknowledgement (ACK). If no ACK is received within a timeout period, it assumes a collision occurred, waits for a randomly chosen time (backoff), and retransmits. The maximum achievable throughput is only about 18.4%.
Slotted ALOHA:
An improvement over Pure ALOHA. Time is divided into fixed equal slots, and a station is only allowed to start transmitting at the beginning of a slot. This halves the vulnerable period and doubles the maximum throughput to approximately 36.8%.





Example:

Imagine two students (Station A and Station B) both trying to speak at the same time in a radio conversation. Their voices overlap and neither is understood. Both stop, wait a random number of seconds, and try again — this time one finishes before the other starts.

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2. CSMA (Carrier Sense Multiple Access)

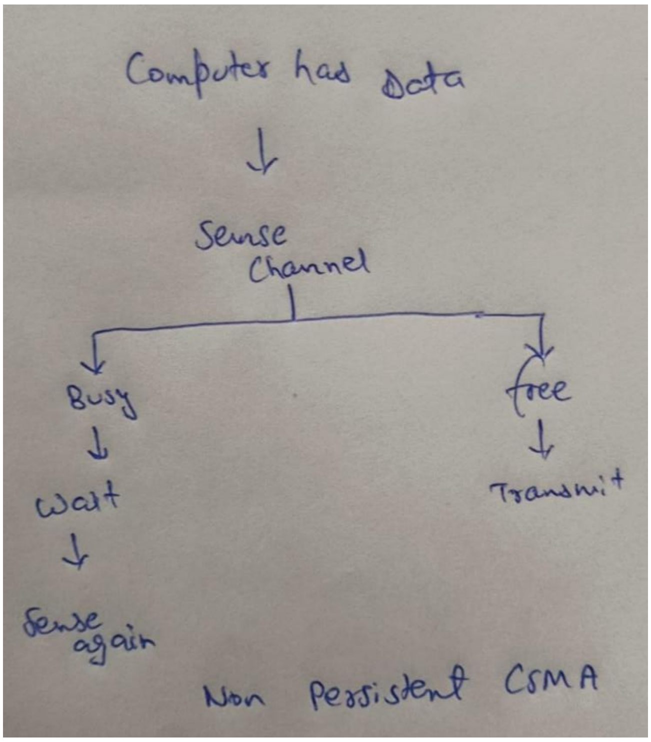
CSMA is a significant improvement over ALOHA. Before transmitting, a station first listens to the channel to check if it is already in use — this is called carrier sensing. Transmission only begins if the channel is detected to be idle, greatly reducing the probability of collisions.

Working:

- Station checks if the channel is idle or busy.
- If idle → begin transmission.
- If busy → wait (strategy depends on persistence type).
- After waiting, sense the channel again and repeat.

Persistence Variants:

- **1-Persistent:** Transmit immediately once channel becomes free. Can still cause collisions if multiple stations were waiting.
- **Non-Persistent:** If busy, wait a random time before sensing again. Reduces collisions but wastes channel time.
- **p-Persistent:** If idle, transmit with probability p and defer with probability (1-p). A balance between the two approaches above.



Example:

Think of a walkie-talkie conversation. Before pressing the button to speak, you listen first. If someone else is already talking, you wait until they finish. This is exactly how CSMA works — listen before you transmit.

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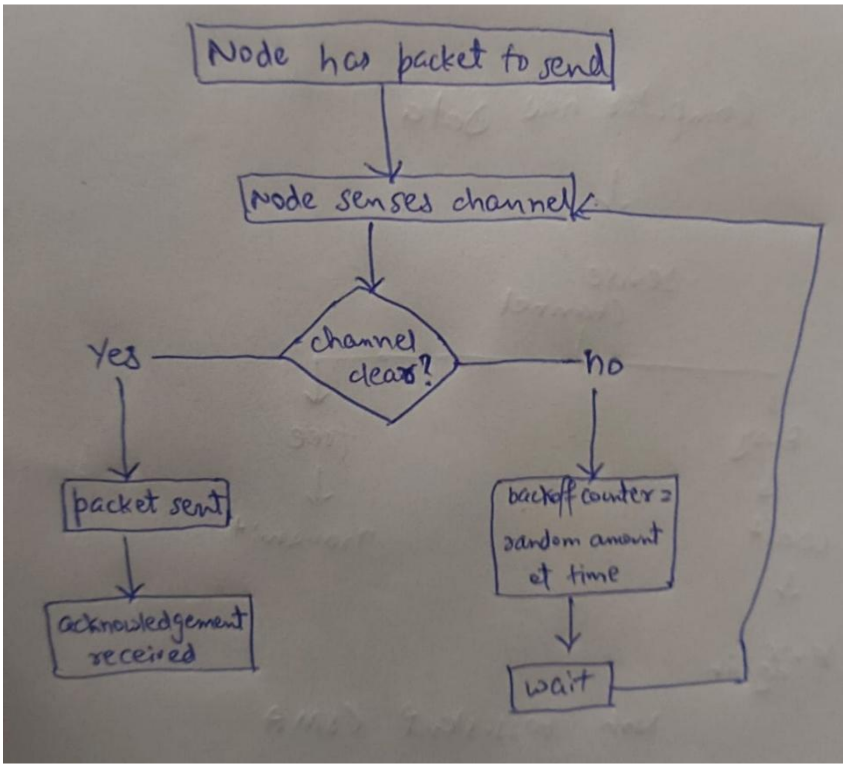
3. CSMA/CD (Carrier Sense Multiple Access with Collision Detection)

CSMA/CD builds on top of CSMA by adding the ability to detect collisions during transmission itself — not just before. It is the protocol underlying traditional wired Ethernet (IEEE 802.3). Since wired networks can electrically detect voltage changes that indicate two simultaneous transmissions, collision detection is feasible.

Working:

1. Station senses the channel before transmitting.
2. If the channel is free, transmission begins.
3. While transmitting, the station continues to monitor the channel.

4. If a collision is detected mid-transmission:
   - Immediately abort the current transmission.
   - Broadcast a Jam Signal (32-bit) to ensure all stations are aware of the collision.
   - Apply Binary Exponential Backoff — wait for a random time interval and retry.
5. If the same frame fails after 16 attempts, the transmission is abandoned and an error is reported.



Example:
Two cars enter a one-lane tunnel from opposite ends simultaneously. Both realize they are on a collision course, reverse out immediately, wait a random amount of time, and then one of them enters first while the other waits.
Application: Wired Ethernet — IEEE 802.3

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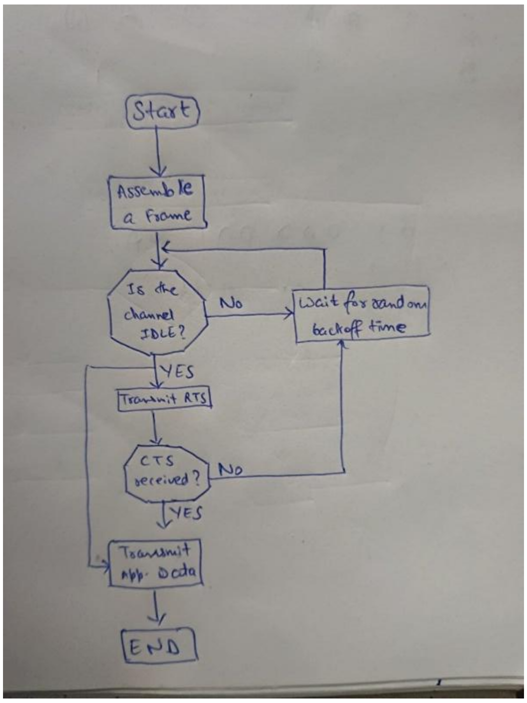
4. CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)
CSMA/CA is designed specifically for wireless networks (IEEE 802.11 — Wi-Fi) where collision detection is not physically possible. In wireless environments, a transmitting station cannot simultaneously listen for collisions on the same frequency it is broadcasting on. Therefore, the strategy shifts from detecting collisions to avoiding them entirely before they happen.
Working:
1. Station senses the channel and waits for it to be idle for a DIFS (Distributed Interframe Space) duration.
2. Once idle, the station sends an RTS (Request to Send) frame to the receiver.
3. The receiver responds with a CTS (Clear to Send) frame — all nearby stations hear this and hold off from transmitting (virtual carrier sensing).
4. The sender transmits the data frame.
5. The receiver sends an ACK to confirm successful receipt.
6. The channel is then released for others to use.



Example:
You want to ask a question in a crowded classroom. Instead of just blurting it out (and clashing with someone else doing the same), you raise your hand first (RTS). The teacher acknowledges you (CTS). Only then do you speak. No clash occurs.
Application: Wireless Networks — IEEE 802.11

Difference between CSMA/CD and CSMA/CA:

Parameter	**CSMA/CD**	**CSMA/CA**
Full Form	Carrier Sense Multiple Access with Collision Detection	Carrier Sense Multiple Access with Collision Avoidance
Core Idea	Collisions are detected after they occur	Collisions are prevented before they occur
Medium	Wired (Ethernet cable)	Wireless (Radio frequency)
IEEE Standard	802.3	802.11
Collision Detection	Possible — voltage change detected on wire	Not possible — station cannot hear while transmitting
Jam Signal	Sent after collision is detected	Not used
RTS/CTS Handshake	Not required	Used to reserve the channel
ACK Frame	Not used	Mandatory after every data frame
Backoff Method	Binary Exponential Backoff after collision	Random backoff before transmission (after DIFS)
Network Efficiency	Higher in low-traffic wired LANs	Slightly lower due to RTS/CTS/ACK overhead
Real-world Example	Ethernet LAN in labs/offices	Wi-Fi in homes/cafes/campus